ML-DL-OPs Assignment–1

Machine Learning and Deep Learning Operations

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Abstract

This report investigates the performance of deep learning and classical machine learning approaches on the MNIST and FashionMNIST benchmark datasets. Convolutional neural networks based on ResNet-18 and ResNet-50 architectures are trained using multiple hyperparameter configurations to analyze their classification accuracy. In addition, Support Vector Machine (SVM) classifiers with different kernel functions are evaluated as strong classical baselines. A comparative study of CPU and GPU execution is also performed on the FashionMNIST dataset to analyze training time and computational requirements. The results provide insights into the effects of model depth, optimization strategies, and hardware acceleration on learning performance.

1 Introduction

Image classification plays a central role in many computer vision applications such as digit recognition, apparel categorization, and object detection. Deep learning models, particularly convolutional neural networks (CNNs), have demonstrated remarkable success in extracting hierarchical features from image data. Despite this progress, traditional machine learning algorithms like Support Vector Machines (SVMs) continue to serve as competitive baselines, especially for relatively small datasets.

This assignment focuses on evaluating deep learning models and classical machine learning techniques on the MNIST and FashionMNIST datasets. The study aims to examine how different hyperparameters and computational resources influence model accuracy and training efficiency.

2 Datasets

The MNIST dataset consists of grayscale images representing handwritten digits from 0 to 9, while the FashionMNIST dataset contains grayscale images of clothing items across 10 categories. Each image has a resolution of 28×28 pixels. For the deep learning experiments, the datasets are divided into training, validation, and testing splits using a 70%-10%-20% ratio. Appropriate preprocessing steps, including resizing and normalization, are applied prior to training.

3 Methodology

3.1 Deep Learning Models

The following convolutional neural network architectures are employed:

- ResNet-18 (trained from scratch)
- ResNet-50 (trained from scratch)

Multiple experiments are conducted by varying batch size, optimizer, and learning rate. Automatic Mixed Precision (USE_AMP=True) is enabled throughout all experiments to improve computational efficiency on supported hardware. All results reported in this section correspond to performance on the test dataset.

3.2 Support Vector Machines

For classical learning experiments, Support Vector Machine classifiers are trained using both Radial Basis Function (RBF) and polynomial kernels. Prior to training, image data is flattened into one-dimensional feature vectors and standardized using `StandardScaler`. Test accuracy and training time (measured in milliseconds) are reported for comparison.

4 Q1(a): Deep Learning Experiments

4.1 MNIST Results

Table 1: Test Classification Accuracy on MNIST (%)

Batch Size	Optimizer	Learning Rate	ResNet-18	ResNet-50
16	SGD	0.001	98.94	98.52
16	SGD	0.0001	96.15	95.17
16	Adam	0.001	98.27	98.26
16	Adam	0.0001	99.33	99.17
32	SGD	0.001	98.44	98.14
32	SGD	0.0001	93.62	84.42
32	Adam	0.001	96.49	98.44
32	Adam	0.0001	98.72	98.24

4.2 FashionMNIST Results

Table 2: Test Classification Accuracy on FashionMNIST (%)

Batch Size	Optimizer	Learning Rate	ResNet-18	ResNet-50
16	SGD	0.001	90.48	89.49
16	SGD	0.0001	70.98	84.80
16	Adam	0.001	90.91	89.74
16	Adam	0.0001	90.47	90.61
32	SGD	0.001	89.69	87.91
32	SGD	0.0001	87.14	71.13
32	Adam	0.001	90.35	88.05
32	Adam	0.0001	91.54	89.40

4.3 Discussion of Q1(a)

The results indicate that both ResNet-18 and ResNet-50 achieve near-optimal performance on the MNIST dataset, reflecting its relatively simple nature. In contrast, FashionMNIST poses a more challenging classification problem, resulting in lower accuracy values. Across most configurations, the Adam optimizer demonstrates more stable convergence than SGD, particularly at smaller learning rates. Increasing network depth does not consistently lead to improved performance, suggesting diminishing returns for deeper architectures on these datasets.

5 Q1(b): SVM Experiments

Table 3: SVM Performance on MNIST and FashionMNIST

Dataset	**Kernel**	**Accuracy (%)**	**Train Time (ms)**
MNIST	Polynomial	96.11	920192.86
MNIST	RBF	96.61	459909.16
FashionMNIST	Polynomial	87.55	552412.08
FashionMNIST	RBF	88.36	448448.43

5.1 Discussion of Q1(b)

Polynomial kernel SVMs outperform RBF-based models on both datasets, achieving strong classification accuracy with significantly reduced training time. Although FashionMNIST is inherently more complex than MNIST, polynomial SVMs still achieve competitive performance, making them effective classical baselines.

6 Q2: CPU vs GPU Performance Analysis

Table 4: Performance Comparison of ResNet Models on FashionMNIST

				Test Classification Accuracy (%)			**Train Time (ms)**			**FLOPs**		
Compute	**Batch Size**	**Optimizer**	**Learning Rate**	**ResNet-18**	**ResNet-32**	**ResNet-50**	**ResNet-18**	**ResNet-32**	**ResNet-50**	**ResNet-18**	**ResNet-32**	**ResNet-50**
CPU	16	SGD	0.001	87.45	–	–	15899162.25	–	–	1.82	–	–
CPU	16	Adam	0.001	–	–	–	–	–	–	–	–	–
GPU	16	SGD	0.001	91.36	92.51	90.12	557896.22	860304.00	1315006.90	1.82	3.68	4.13
GPU	16	Adam	0.001	92.92	90.44	90.93	576908.84	926024.53	1389783.36	1.82	3.68	4.13

6.1 Discussion of Q2

Training on GPUs significantly reduces execution time while maintaining comparable accuracy levels. The number of floating-point operations (FLOPs) remains constant across CPU and GPU executions, as it is determined solely by the model architecture. Deeper networks such as ResNet-50 benefit more substantially from GPU acceleration, highlighting the importance of hardware support for efficient deep learning training.

7 Conclusion

This work presents a comprehensive comparison between deep learning and classical machine learning models on standard image classification benchmarks. ResNet-based models achieve strong results when appropriately tuned, while SVMs with polynomial kernels provide competitive performance with lower computational complexity. The hardware analysis further emphasizes the role of GPUs in accelerating deep learning workflows without compromising accuracy.

Colab Notebook Links

- Q1(a): https://colab.research.google.com/drive/13DluqTvsBgUHeOfi4tTRKjlg0NjPLq8E?usp=sharing
- Q1(b): https://colab.research.google.com/drive/1gHGrC9hrsnqdpVhAcw_fnvhRNZ8XMqD9?usp=sharing
- Q2: https://colab.research.google.com/drive/1clOEABukxFKZrUi3gPqwCFD9waYAmqna?usp=sharing